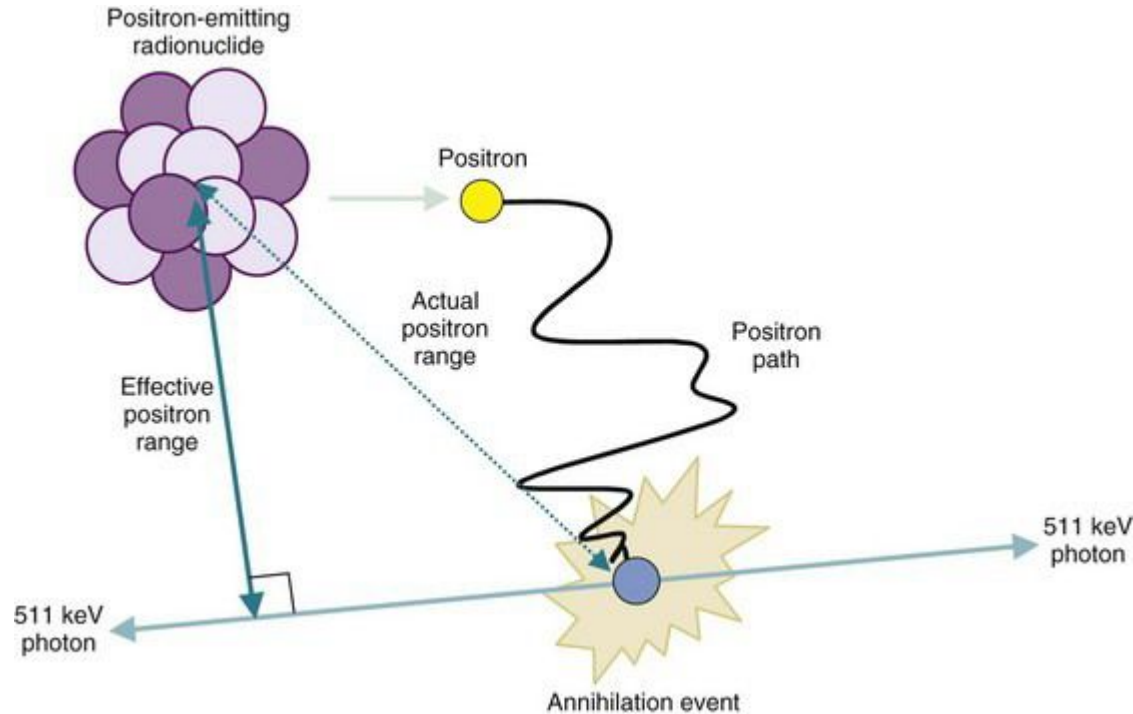


Positron range impact on PET imaging

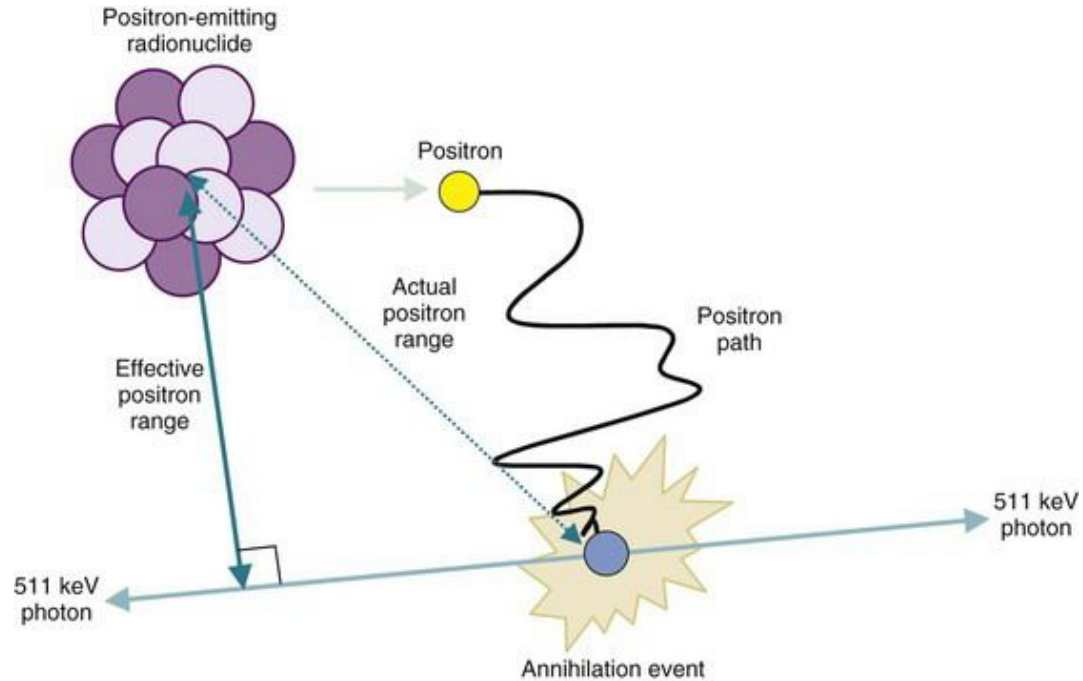
Szymon Niedźwiecki

Plan of presentation

1. Motivation
2. Isotopes and eject energies
3. Interaction of positrons with matter
4. Estimation of positron range
5. State of the art
6. Summary

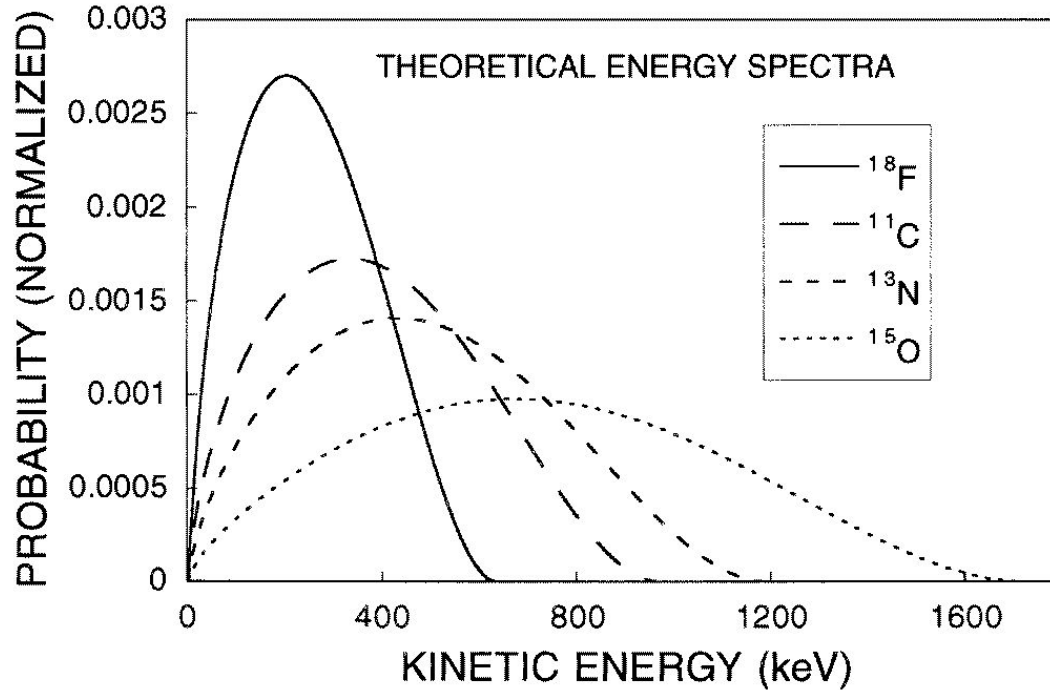


Motivation



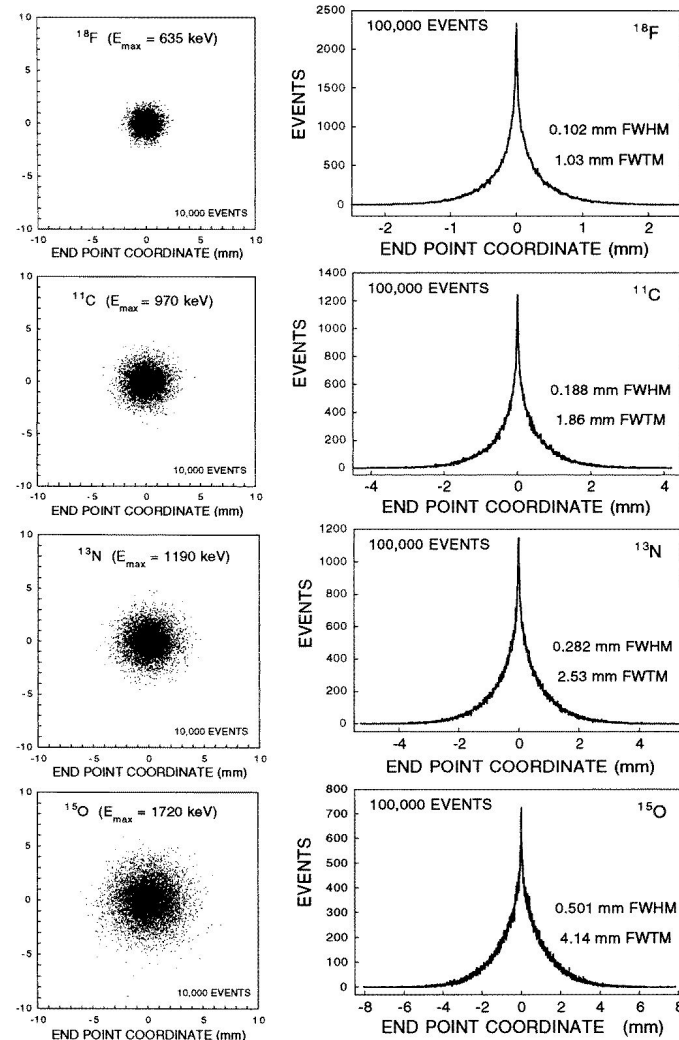
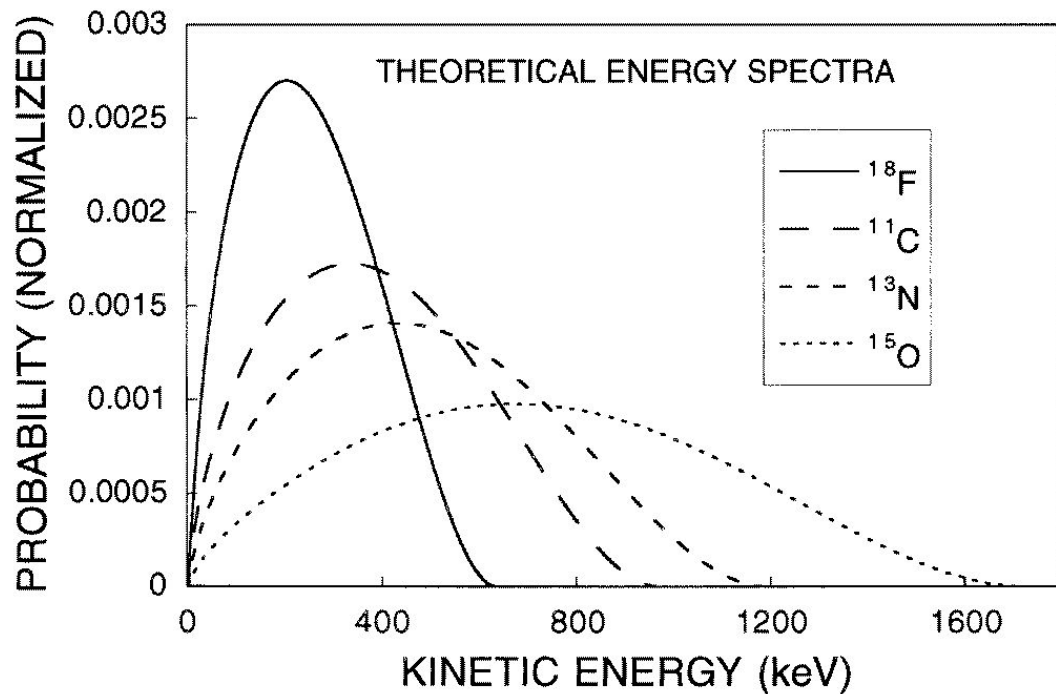
Motivation

Craig S Levin and Edward J Hoffman 1999 Phys. Med. Biol. 44
781



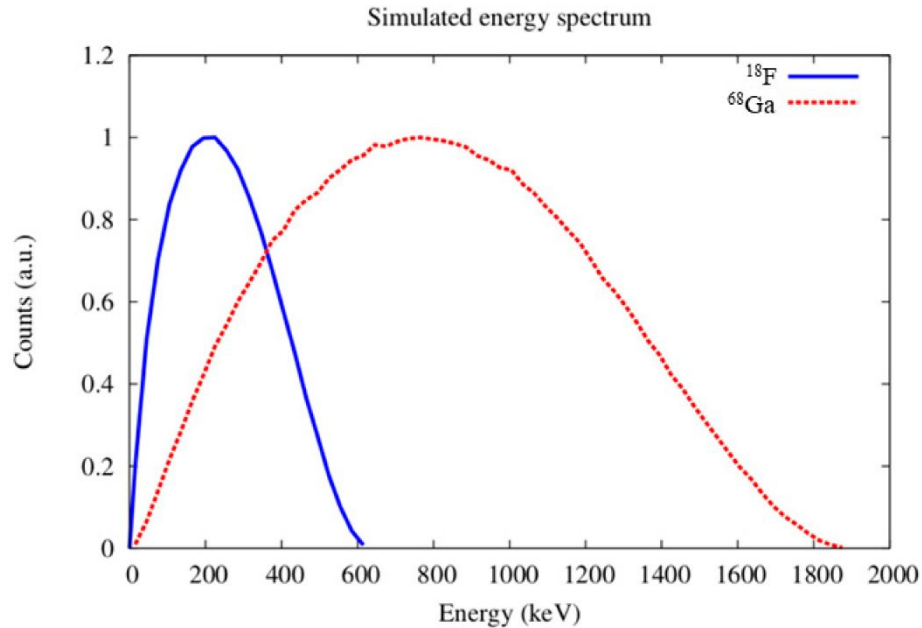
Motivation

Craig S Levin and Edward J Hoffman 1999 Phys. Med. Biol. 44 781



Motivation

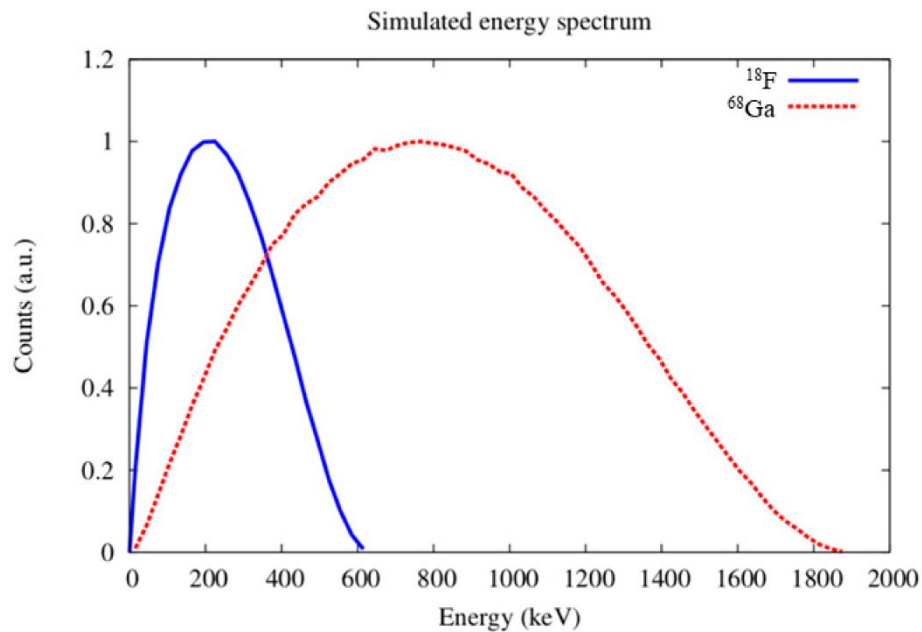
Herraiz, J.L.; Bembibre, A.; López-Montes, A. Deep-Learning Based Positron Range Correction of PET Images. Appl. Sci. 2021, 11, 266. <https://doi.org/10.3390/app11010266>



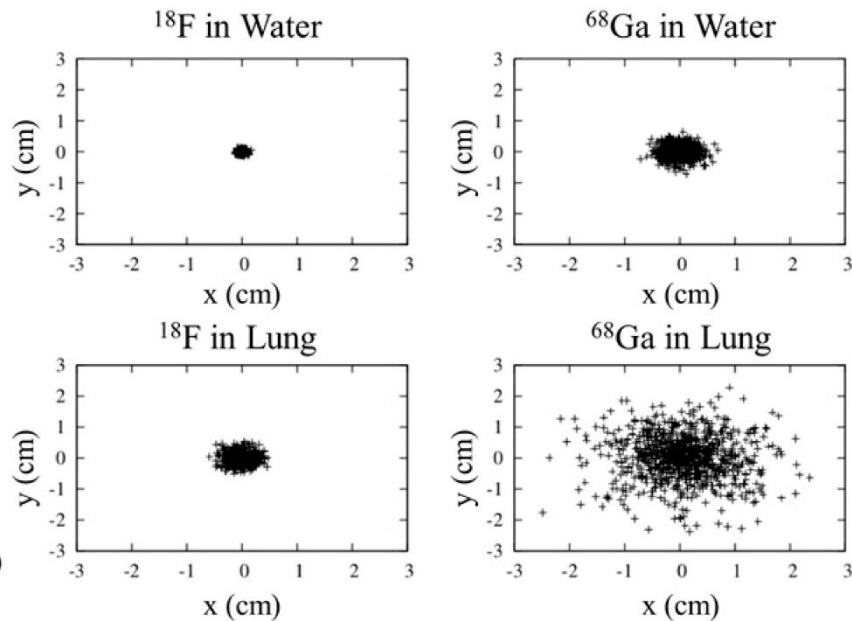
(a)

Motivation

Herraiz, J.L.; Bembibre, A.; López-Montes, A. Deep-Learning Based Positron Range Correction of PET Images. Appl. Sci. 2021, 11, 266. <https://doi.org/10.3390/app11010266>



(a)



(b)

Isotopes in PET and positron energies

Isotope	β^+ Rate [%]	Max β^+ Energy [MeV]	Mean β^+ Energy [MeV]	Mean β^+ Range in Tissue [mm]	Prompt Energy [keV]	Prompt Intensity [%]	Half-life
F-18	96.7	0.6335	~0.25	0.6	–	–	1.83 h

Kemerink, G.J., Visser, M.G.W., Franssen, R. et al. Effect of the positron range of ^{18}F , ^{68}Ga and ^{124}I on PET/CT in lung-equivalent materials. Eur J Nucl Med Mol Imaging 38, 940–948 (2011).

IAEA Nuclear Data Section – Decay data for positron-emitting isotopes (endpoint energies, gamma intensities).

Rosar T et al. Performance comparison of Sc-44 to Ga-68 using preclinical PET scanners. Journal of Nuclear Medicine.

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Ga-68	89	1.899	0.83	3.6	1077	3.0	1.13 h
I-124	0.30 / 10.8	0.812 / 2.138	~1.81 (weighted)	3.48	603 / 723 / 1691	62.9 / 10.35 / 10.88	4.18 d

Kemerink, G.J., Visser, M.G.W., Franssen, R. et al. Effect of the positron range of 18F, 68Ga and 124I on PET/CT in lung-equivalent materials. Eur J Nucl Med Mol Imaging 38, 940–948 (2011).

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Sc-44	94.3	1.471	0.63	2.5	1157	>99.9	4.04 h

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F-18	96.7	0.6335	~0.25	0.6	2.4
Ga-68	89	1.899	0.83	3.6	8.9
I-124	0.30 / 10.8	0.812 / 2.138	~1.81 (weighted)	3.48	6.9-10.2
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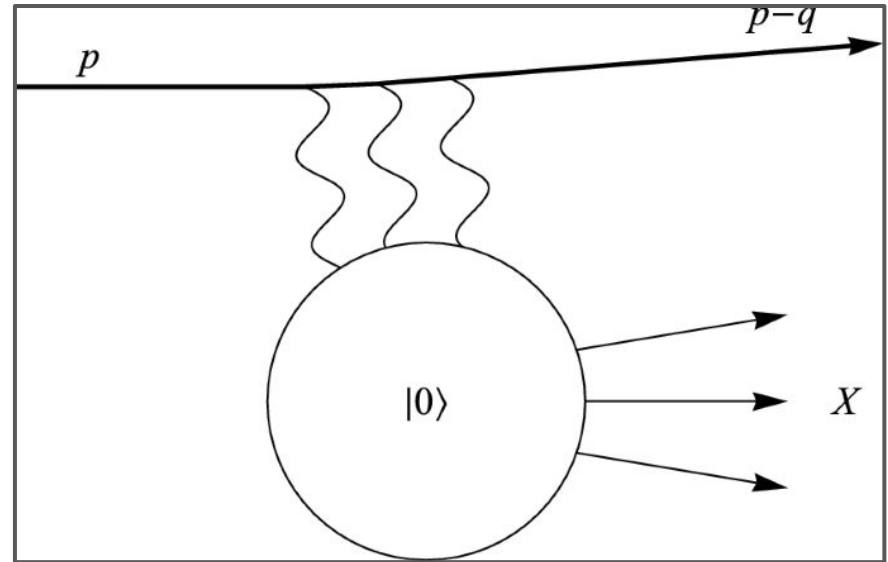
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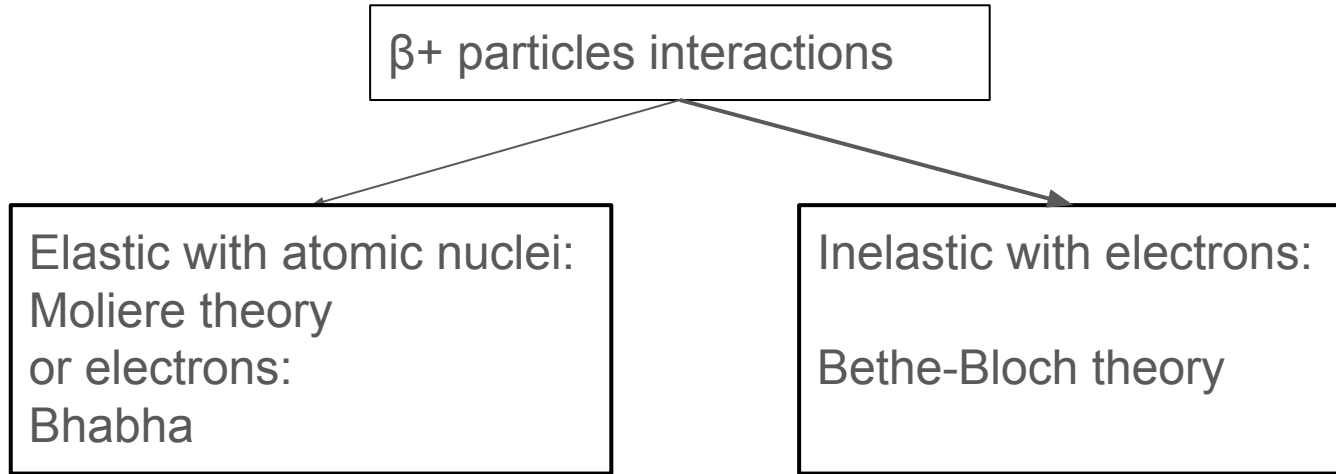
Interactions of positrons with matter

β^+ particles interactions

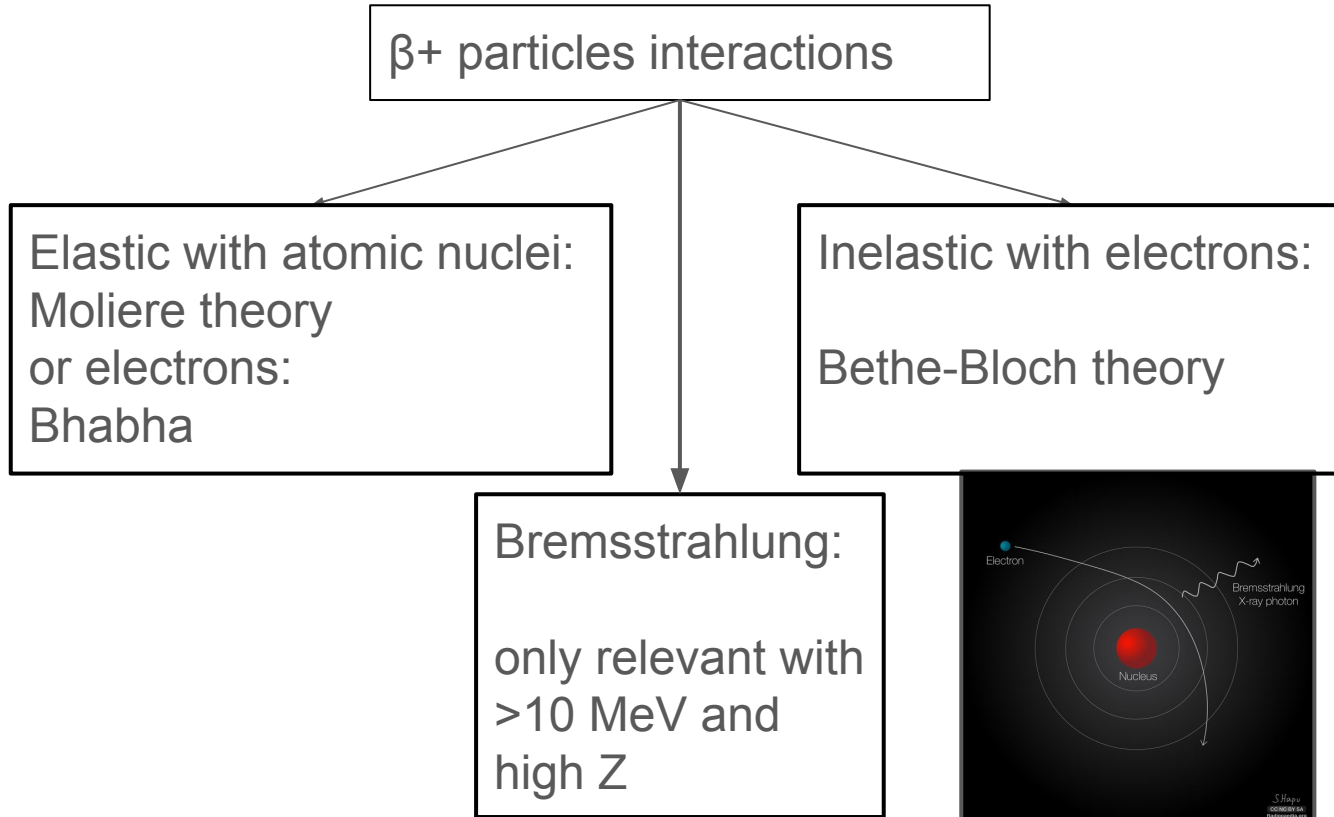
Elastic with atomic nuclei:
Moliere theory
or electrons:
Bhabha



Interactions of positrons with matter

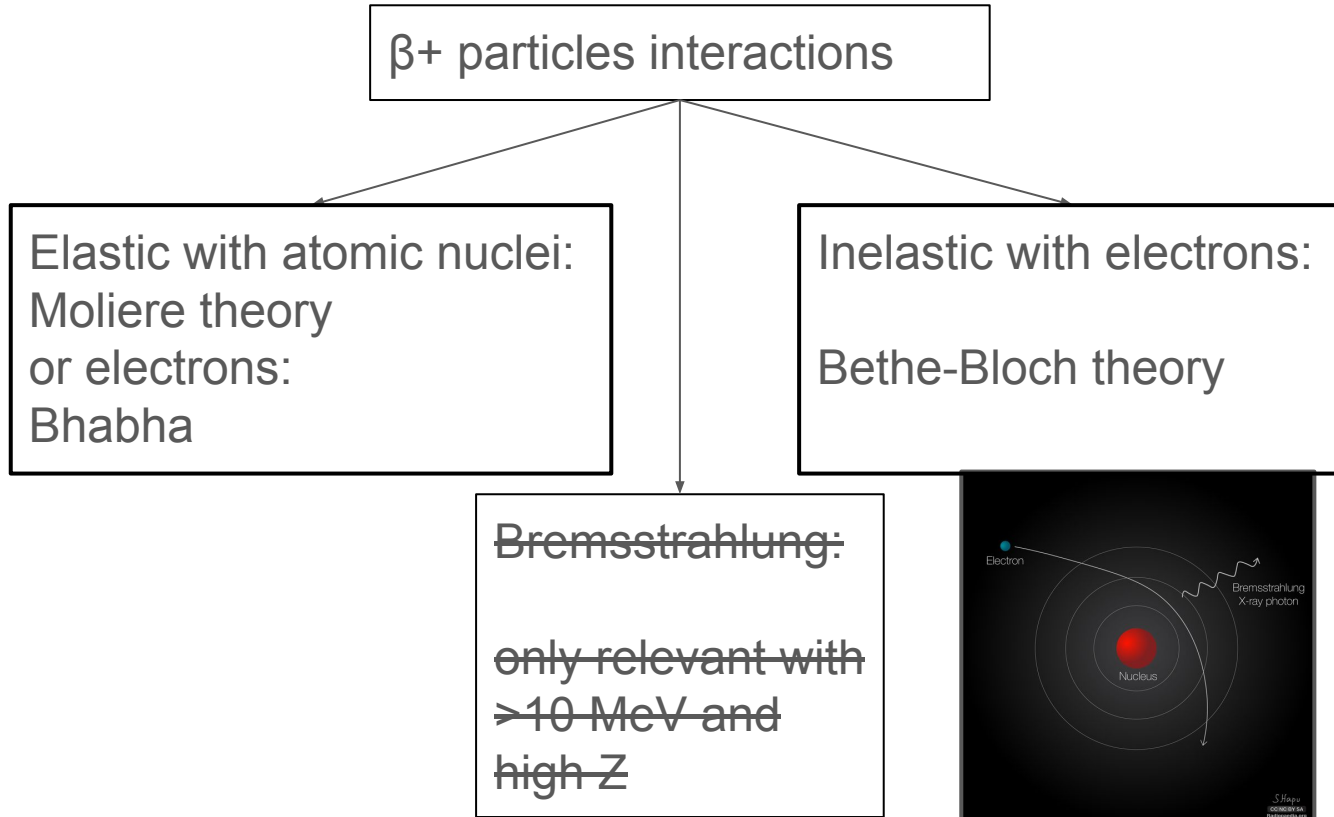


Interactions of positrons with matter



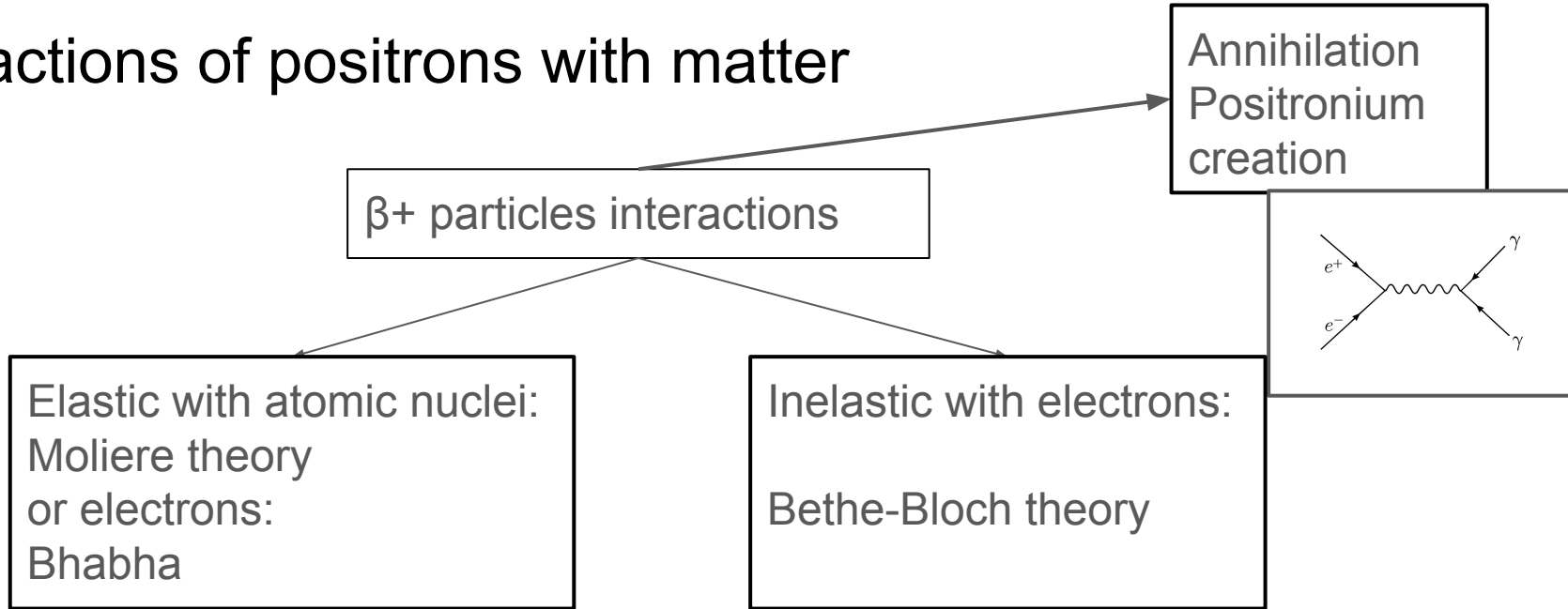
<https://radiopaedia.org/articles/bremsstrahlung-radiation>

Interactions of positrons with matter

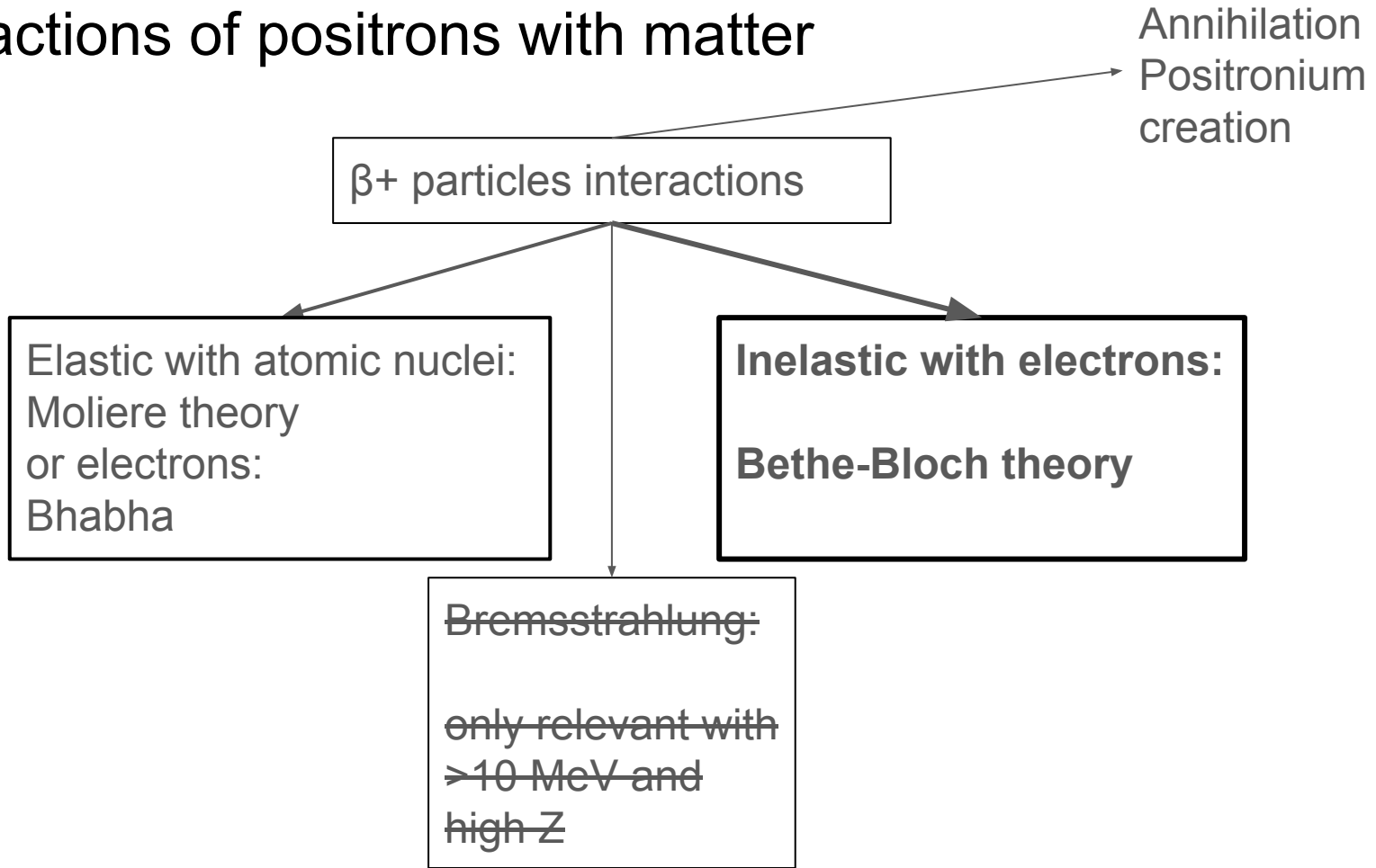


<https://radiopaedia.org/articles/bremsstrahlung-radiation>

Interactions of positrons with matter



Interactions of positrons with matter



Estimation of positron range

Model	Description	Energy range	Comments
Bethe-Bloch	Energy loss due to Coulomb interactions with charged particles	~0.1 - few MeV	Annihilation is not included

$$\frac{dE}{dx} = 4\pi r_0^2 \frac{mc^2}{\beta^2} N Z (A + B)$$

$$A = \ln \left(\frac{\beta \gamma \sqrt{\gamma - 1} mc^2}{I} \right)$$

$$B = \frac{1}{2\gamma^2} \left(\frac{(\gamma - 1)^2}{8} + 1 - (2\gamma^2 + 2\gamma - 1) \ln 2 \right)$$

Estimation of positron range

Model	Description	Energy range	Comments
Bethe-Bloch	Energy loss due to Coulomb interactions with charged particles	~0.1 - few MeV	Annihilation is not included
CSDA (Continuous Slowing Down Approximation)	Energy is lost continuously, no scattering	keV - MeV	Gives upper limit on range

$$R(E_0) = \int_{E_T}^{E_0} \frac{dE}{(S/\rho)}$$

$$(S/\rho)_{comp.} = w_1 \left(\frac{S_1(E)}{\rho} \right) + w_2 \left(\frac{S_2(E)}{\rho} \right) + \dots$$

$$S_{\xi} = - \frac{dE}{d\xi} = \frac{dE}{\rho dx} = \frac{4\pi}{mv^2} \frac{e^4}{A} \frac{Z^{*2}}{Z_2^*} \left\{ \ln \left(\frac{E}{I^*} \right) - F(\tau) \right\} \quad (1)$$

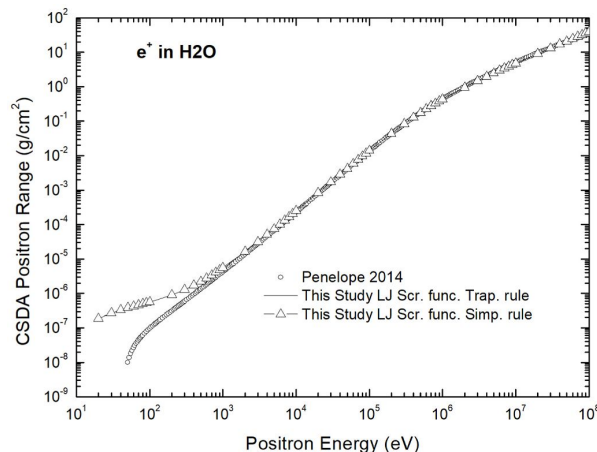
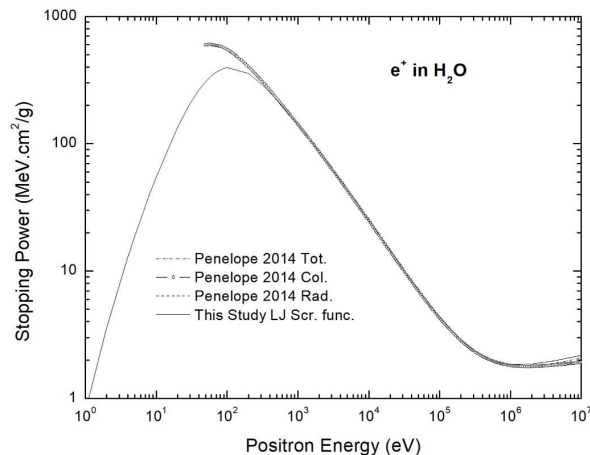
where

$$F^+(\tau) = 2\ln(2) - \beta \left/ 12 \left[23 + \frac{14}{(\tau+2)} + \frac{14}{(\tau+2)^2} + \frac{14}{(\tau+2)^3} \right] \right. \text{ for positrons} \quad (2)$$

$$\frac{4\pi}{mv_1^2} \frac{e^4}{A} \frac{Z^{*2}}{Z_2^*} = (k / A\beta^2) z^{*2} Z_2^*, \quad (3)$$

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Radiation Physics and Chemistry 196 (2022)
110092, H. Gümüş,

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Quantum corrections	Includes trapping, annihilation and diffusion	eV - keV	Important in solid state physics

CSDA vs MC

Isotope	CSDA (Rmax)	MC (Rmean)	Remarks
F-18	~2.4 mm	~0.6 mm	CSDA overestimates by ~4×
Ga-68	~8.9 mm	~3.6 mm	Large difference, impacts resolution
Sc-44	no CSDA data	~2.5 mm	MC from phantom studies
I-124	~10 mm	~3.5 mm	MC shows long tails in distribution

Note:
depending on
the method a
different “range”
is studied

F-18

Levin CS, Hoffman EJ. Calculation of positron range and its effect on PET resolution. Phys Med Biol. 1976;21:113-21.
Hindié E et al. Positron range in PET imaging: isotope comparison. EJNMMI Physics, 2024.

Ga-68

Van Dalen JA et al. Impact of positron range on PET resolution. J Nucl Med, 2008.
Hindié E et al. Monte Carlo simulations for PET isotopes. EJNMMI Physics, 2024.

Sc-44

Rosar T et al. Performance comparison of Sc-44 to Ga-68 using preclinical PET scanners. J Nucl Med, 2020.

I-124

Van Dalen JA et al. Impact of positron range on PET resolution. J Nucl Med, 2008.
Alanazi M et al. Effect of positron range on PET image quality. BJSTR, 2022.

... vs experiment

The Impact of Positron Range on PET Resolution, Evaluated with Phantoms and PHITS Monte Carlo Simulations for Conventional and Non-conventional Radionuclides

L. M. Carter¹, Adam Leon Kesner², E. C. Pratt^{1,3}, V. A. Sanders⁴, A. V. F. Massicano⁵, C. S. Cutler⁴, S. E. Lapi⁵, Jason S. Lewis^{1,3,6,7,8}

$$I_{PET} = I_{RT} \otimes PSF_{sys}$$

MC: PHITS
studied on phantoms:
Jaszczak
Small animal

... vs experiment

**The Impact of Positron Range on PET Resolution, Evaluated with
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Small animal

$$I_{PET} = I_{RT} \otimes PSF_{sys} = I_{RT} \otimes aPSF \otimes PSF_{scanner}$$

... vs experiment

Properties of PET radionuclides examined

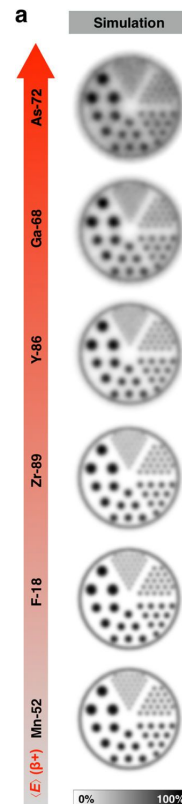
Radionuclide	Half-life	$\langle E_{\beta^+} \rangle$ (keV)	CDSA range [*] (cm)	Advantages
⁷² As	26.0 h	1170	0.516	• Intermediate half-life • Generator-based production • High positron intensity (87.8 %) • Theranostic complement (⁷⁷As)
⁶⁸ Ga	68 min	830	0.337	• Generator-based production • High positron intensity (88.9 %) • Theranostic twins (¹⁷⁷Lu, ⁹⁰Y) • Favorable dosimetric profile for pretargeted immunoPET
⁸⁶ Y	14.7 h	660	0.250	• Theranostic isotopologue (⁹⁰Y)
⁸⁹ Zr	78.4 h	396	0.123	• Optimal half-life for traditional immunoPET • High resolution
¹⁸ F	110 min	250	0.062	• High resolution • High positron intensity (96.7 %) • Favorable dosimetric profile • For pretargeted immunoPET
⁵² Mn	5.59 days	242	0.059	• High-resolution immunoPET

* Range taken in the continuous slowing down approximation (CDSA) in water for the average energy positron emission (data obtained from NIST);

... vs experiment

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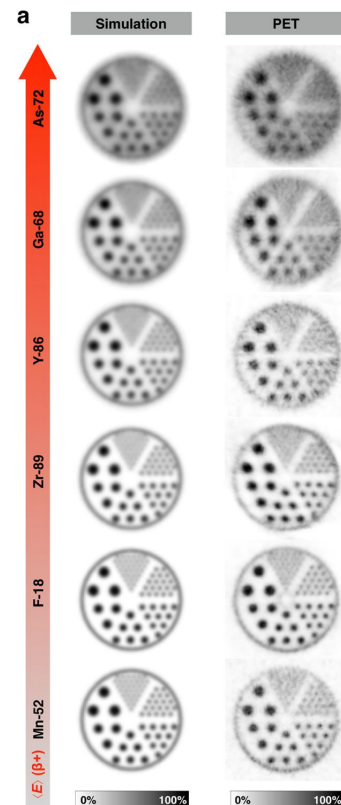
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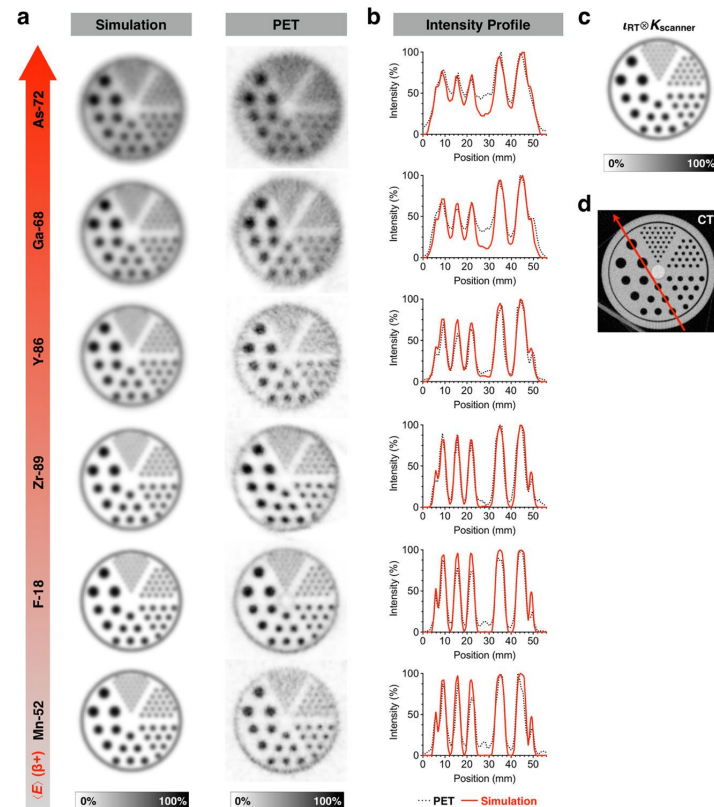


... vs experiment

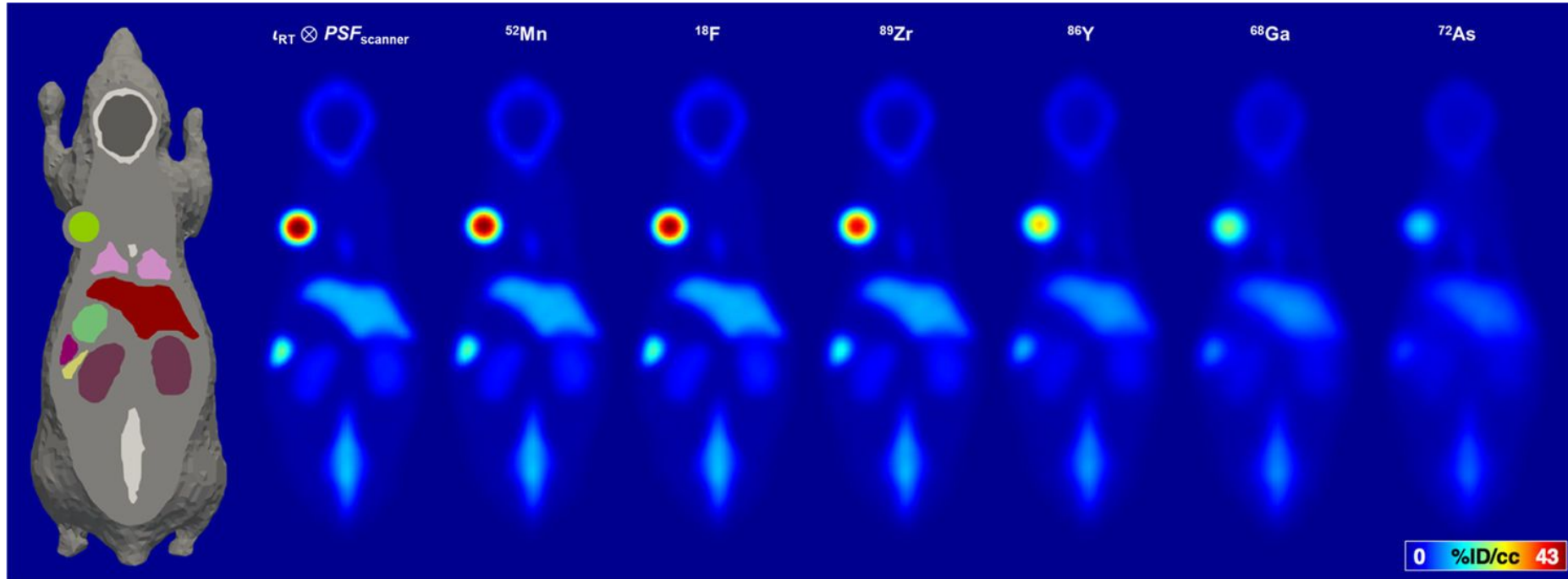
Properties of PET radionuclides examined

Radionuclide	Half-life	$\langle E_{\beta^+} \rangle$ (keV)	CDSA range [*] (cm)	Advantages
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^{68}Ga	68 min	830	0.337	- Generator-based production - High positron intensity (88.9 %) - Theranostic twins (^{177}Lu, ^{90}Y) - Favorable dosimetric profile for pretargeted immunoPET
^{86}Y	14.7 h	660	0.250	Theranostic isotopologue (^{90}Y)
^{89}Zr	78.4 h	396	0.123	- Optimal half-life for traditional immunoPET - High resolution
^{18}F	110 min	250	0.062	- High resolution - High positron intensity (96.7 %) - Favorable dosimetric profile - For pretargeted immunoPET
^{52}Mn	5.59 days	242	0.059	High-resolution immunoPET

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... vs experiment



Simulation!

... vs experiment

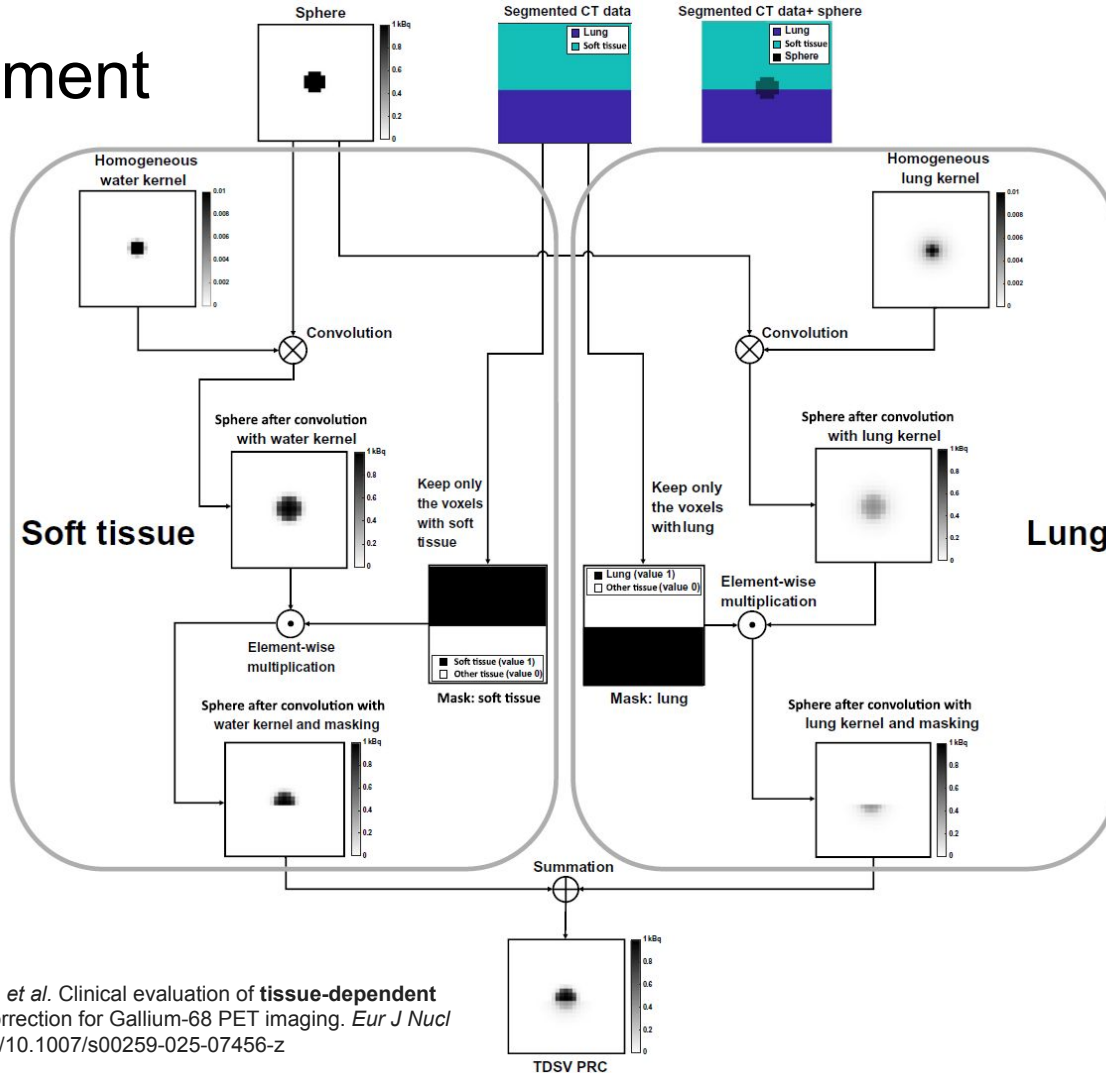
Clinical evaluation of tissue-dependent and spatially-variant positron range correction for Gallium-68 PET imaging

Prodromos Gavriilidis^{1,2,3} · Michel Koole³ · Felix M. Mottaghy^{1,4} · Floris P. Jansen⁵ · Roel Wiertsema¹ 

MC: GATE

Studied on patients

... vs experiment



... vs experiment

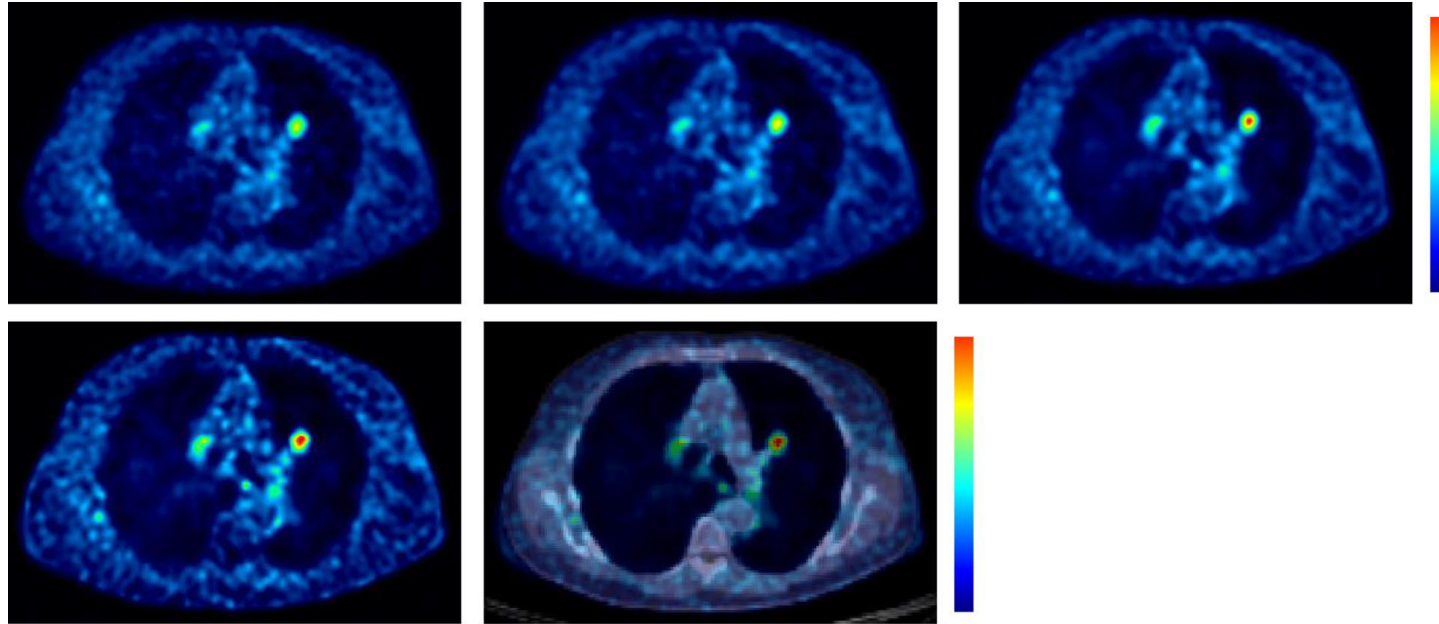


Fig. 10 Image comparison for lung lesion. (a) Non-PRC. (b) Tissue-independent PRC. (c) TDSV PRC with 6.0 mm FWHM filter. (d) TDSV PRC with 4.0 mm FWHM filter. (e) PET/CT fusion for the TDSV PRC with 4.0 mm FWHM filter

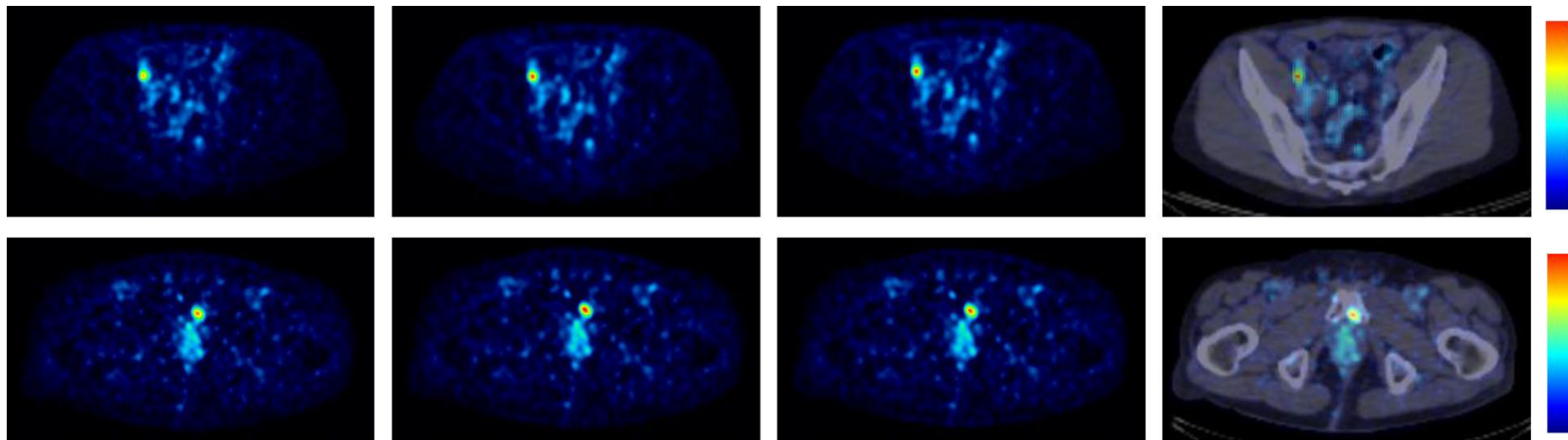


Fig. 8 Image comparison of soft tissue and bone lesions. From left to right, the columns show non-PRC, tissue-independent PRC, TDSV PRC, and PET/CT fusion for the TDSV PRC. Soft tissue lesion (lymph node): (a)-(d). Bone lesion (metastasis): (e)-(h)

Gavrilidis, P., Koole, M., Mottaghy, F.M. *et al.* Clinical evaluation of **tissue-dependent and spatially-variant** positron range correction for Gallium-68 PET imaging. *Eur J Nucl Med Mol Imaging* (2025). <https://doi.org/10.1007/s00259-025-07456-z>

Summary

Positron Range Correction involves two step process:

- MC simulation of positron range;
- application of correction during image reconstruction (via masking);

BUT:

- requires density map of tissues;

Is a natural extension of current reconstruction algorithm.

Positron Range Corrections will have most impact on:

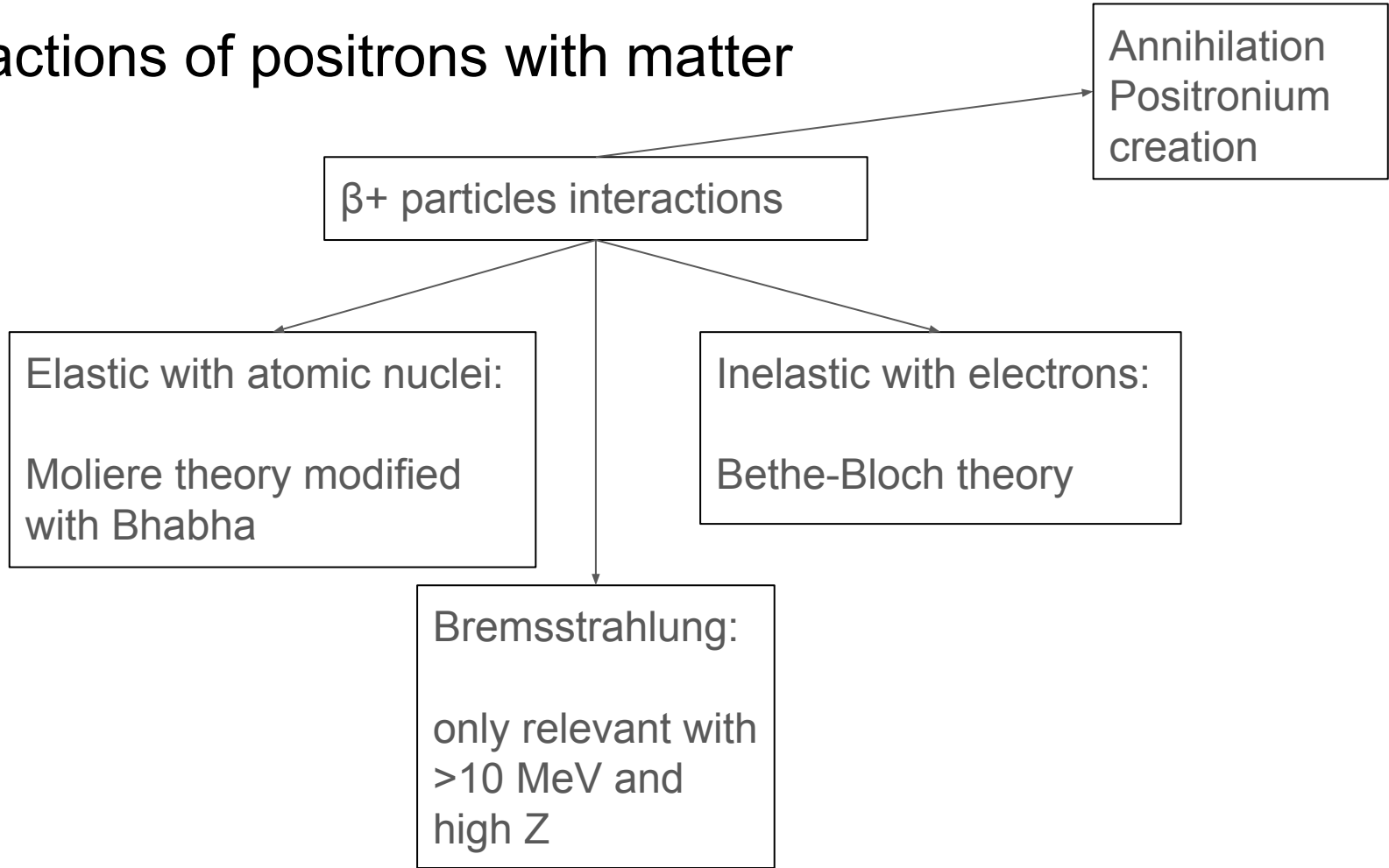
- images produced by Ga-68, Sc-44 and other isotopes with high energies of e^+ ;
- lesions localised between two distinct tissues;
- multi-isotope studies.

Thank you for Your attention!

Questions?

Backup

Interactions of positrons with matter



CSDA - Continuous Slowing Down Approximation

The modified collision SP formula for incoming charged particles can be written as (Gümüş, 2005,2008; Gümüş and Koksall, 2002; Rohrlich and Carlson, 1954; Sugiyama, 1985)

$$S_{\xi} = -\frac{dE}{d\xi} = \frac{dE}{\rho dx} = \frac{4\pi e^4 z^{*2}}{mv^2} \frac{N_0}{A} Z_2^* \left\{ \ln\left(\frac{E}{I^*}\right) - F(\tau) \right\} / 2 \quad (1)$$

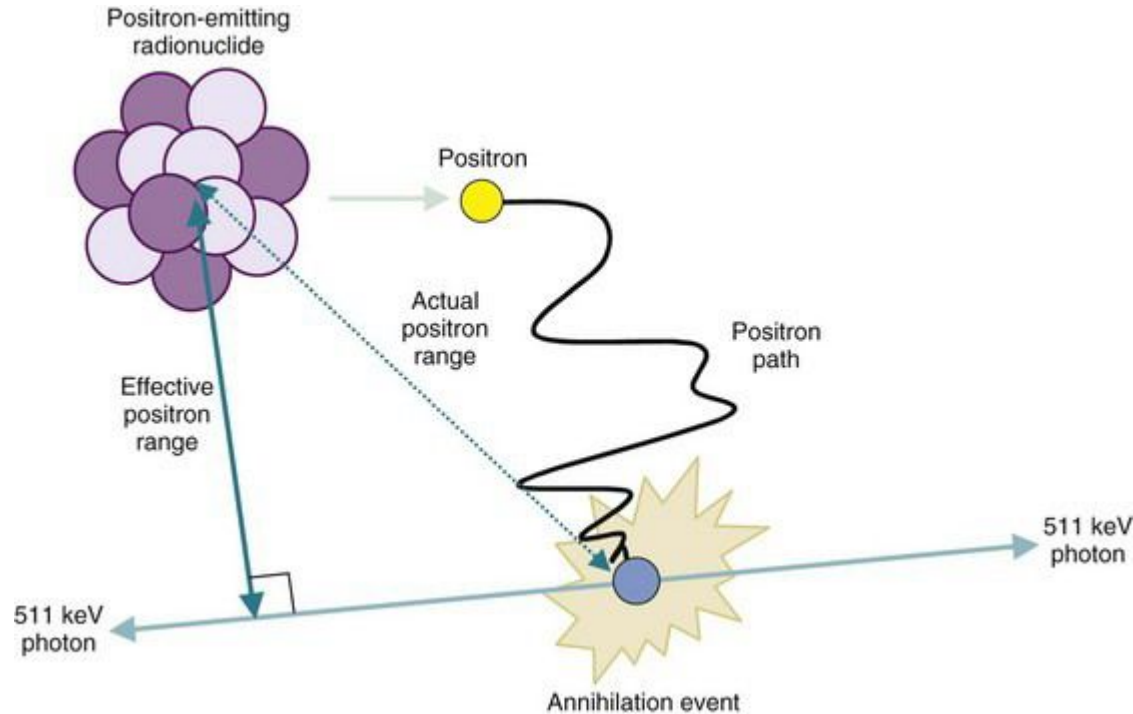
where

$$F^+(\tau) = 2\ln(2) - \beta \left/ 12 \left[23 + \frac{14}{(\tau+2)} + \frac{14}{(\tau+2)^2} + \frac{14}{(\tau+2)^3} \right] \right. \text{ for positrons} \quad (2)$$

$$\frac{4\pi e^4 z^{*2}}{mv_1^2} \frac{N_0}{A} Z_2^* = (k / A\beta^2) z^{*2} Z_2^*, \quad (3)$$

Plan of presentation

1. Motivation
2. Theoretical models
3. Impact on PET
4. Literature studies
5. Models vs experiment
6. What are we missing?
7. Research map
8. My focus
9. Grants?



Theoretical models

Model	Description	Energy range	Comments
Bethe-Bloch	Energy loss due to Coulomb interactions with charged particles	~0.1 - few MeV	Annihilation is not included
CSDA (Continuous Slowing Down Approximation)	Energy is lost continuously, no scattering	keV - MeV	Gives upper limit on range
Kanaya-Okayama	Empirical model for electrons, adapted for positrons	1 - 30 keV	Energy range not useful in PET, only works for specific materials
Monte Carlo (Geant4, Penelope, MCNP etc.)	Particles are tracked with specified steps	eV - GeV	This is PET standard, resource hungry, depends on models applied
Quantum corrections	Includes trapping, annihilation and diffusion	eV - keV	Important in solid state physics

Bethe-Bloch

Beta particles interactions

Elastic with atomic nuclei:

Moliere theory

$$f(\theta)\theta \, d\theta = \vartheta \, d\vartheta (f^{(0)}(\vartheta) + B^{-1}f^{(1)}(\vartheta) + B^{-2}f^{(2)}(\vartheta) + \dots)$$

$$f^{(n)}(\vartheta) = \frac{1}{n!} \int_0^\infty u \, du \, J_0(\vartheta u) \exp\left(-\frac{u^2}{4}\right) \left[\frac{u^2}{4} \ln\left(\frac{u^2}{4}\right)\right]^n$$

$$f^{(0)}(\vartheta) = 2e^{-\vartheta}$$

$$f^{(1)}(\vartheta) = \frac{2(1 - 5\vartheta^{-2})^{-4/5}}{\vartheta^4}$$

$$f^{(2)}(\vartheta) = \frac{16(\ln \vartheta + \ln 0.4)}{\vartheta^6(1 - 9\vartheta^{-2} - 24\vartheta^{-4})}$$

$$\vartheta = \theta/(\chi_c B^{1/2})$$

$$b = B - \ln B$$

$$\chi_c^2 = \frac{4\pi N t e^4 Z(Z+1)}{(p v)^2}$$

Inelastic with electrons:

Bethe-Bloch theory

$$\frac{dE}{dx} = 4\pi r_0^2 \frac{mc^2}{\beta^2} N Z (A + B)$$

$$A = \ln\left(\frac{\beta\gamma\sqrt{\gamma-1}mc^2}{I}\right)$$

$$B = \frac{1}{2\gamma^2} \left(\frac{(\gamma-1)^2}{8} + 1 - (2\gamma^2 + 2\gamma - 1) \ln 2 \right) \quad 45$$